CAMERA MODULE TECHNICAL DATASHEET

Model: [DX3361] **Revision:** [V1.0]

1. Product Overview

PRODUCT IMAGE	
	Model: **DX3361** Image Sensor: **AR0234** Resolution: **2 MP per camera** Interface: **USB 3.0** Frame Rate: **4000 × 1200 @ 60 fps** Module Size: **90 × 24.5 x 31 mm**

2. Key Features

- Dual-camera stereo imaging with synchronized frame capture
- Global Shutter technology for capturing fast-moving scenes
- Up to 60 fps image output
- Integrated 6-axis IMU for motion sensing
- Exposure timestamp support for image and motion data processing
- USB 3.0 interface with driver-free operation on supported UVC-compatible systems
- Fixed-focus optical configuration
- Automatic exposure control (AEC), automatic white balance (AWB) and automatic gain control (AGC)
- Suitable for robotics, stereo vision and visual perception applications

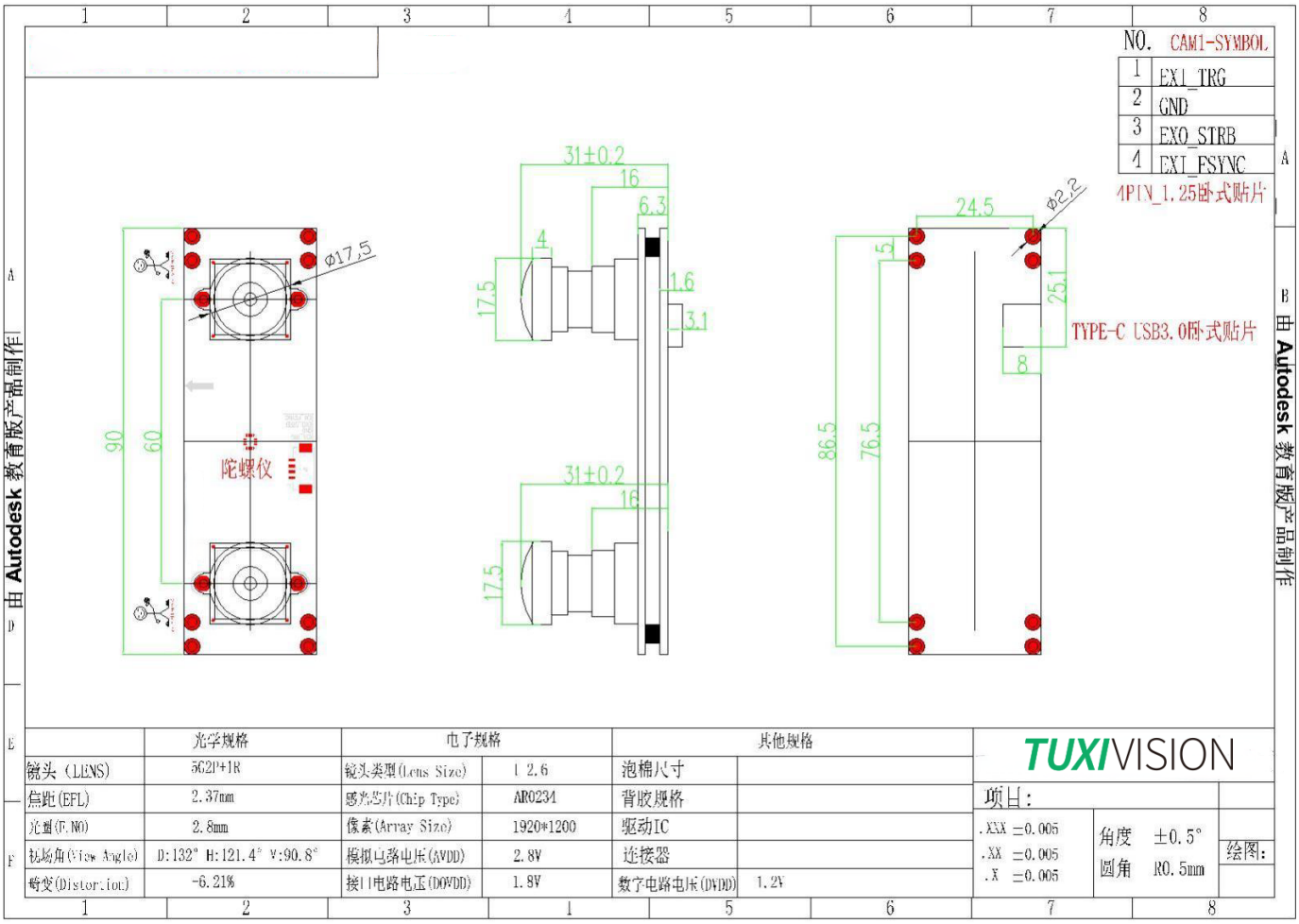
3. Technical Specifications

Item	Specification
Model	USB 3.0 Global Shutter Camera Module
Sensor	AR0234
Sensor Type	CMOS, Global Shutter
Resolution	1920 × 1200 effective pixels
Pixel Size	3.0 μm × 3.0 μm
Video Output	MJPEG / YUY2
Maximum Frame Rate	4000 × 1200 @ 60 fps (MJPEG); 4000 × 1200 @ 30 fps (YUY2)
Lens	1/2.6-inch format
Field of View	100° /130° /160° /175° /195° /200°
Focus	Fixed Focus
Interface	USB 3.0 High-Speed
Operating Temperature	0°C to +60°C
Storage Temperature	-40°C to +80°C
Module Dimensions	90 × 24.5 x 31 mm
OS Compatibility	Windows 10 / Windows 11 / Linux with UVC / macOS X 10.4.8 or later / Android 4.0 or above with UVC

Additional Imaging Parameters

Item	Specification
Maximum Effective Pixels	1920 (H) × 1200 (V)
Center Resolution	800 LW/PH
Signal-to-Noise Ratio (SNR)	37 dB
Dynamic Range	70 dB
Sensitivity	1.4 V/lux·s @ 550 nm
Automatic Exposure Control (AEC)	Supported
Automatic White Balance (AWB)	Supported
Automatic Gain Control (AGC)	Supported
IMU	TDK ICM-42688-P
USB Cable Length	1 m standard; 2 m / 3 m / 5 m optional

4. Mechanical Dimensions



Unit: mm. Dimensions are for reference. Please refer to the latest approved mechanical drawing for production.

Module Dimensions: 90 × 24.5 x 31 mm mm

5. Notes

- Specifications are subject to change without prior notice.
- Actual imaging performance may vary depending on lighting conditions and host platform.
- Operating temperature and stable-image temperature are specified separately.
- The module uses USB bus power and has a specified operating current of 400 mA.
- Handle the module using appropriate electrostatic discharge (ESD) precautions.
- Avoid bending the PCB or FPC during storage, transportation, assembly, disassembly, and testing.
- Do not connect or disconnect the module while powered on.
- Ensure that the cable is connected in the correct orientation and pin alignment.
- Please contact TUXIVISION for the latest specification and mechanical drawing.